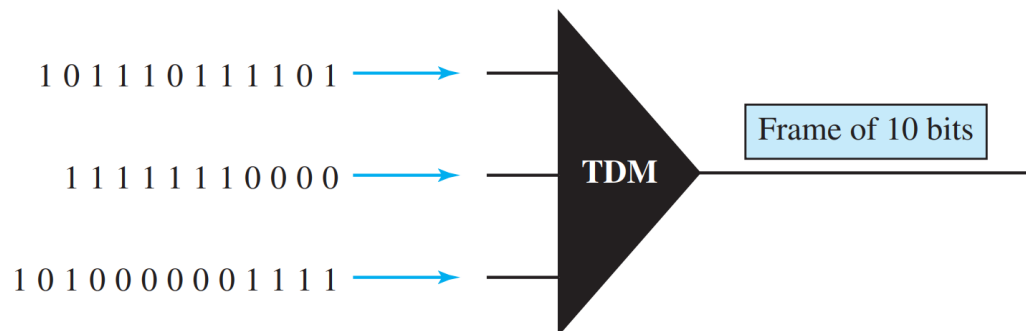


CSE 320 Quiz 3

Marks - 15 Time - 25 min

ID	
Name	

1. In an FDM system, five signals are multiplexed together with a guard band between each signal. If the bandwidth of each signal is 2 kHz, and the guard band is 500 Hz, calculate the total bandwidth required for the FDM transmission. **[2]**
2. Explain the difference between multilevel TDM, multiple-slot TDM, and pulse-stuffed TDM. **[3]**
3. The figure below shows a multiplexer in a synchronous TDM system. Each output slot is only **10 bits** long (3 bits taken from each input plus 1 framing bit). What is the output stream? Distinguish the frames. The bits arrive at the multiplexer as shown by the arrows. **[5]**
[Use 1 as framing bit]



4. An **FHSS** system uses a **5-bit** PN sequence. If the bit rate of the PN is **64 bits per second**, what is the time needed to finish a complete cycle of PN? **[5]**